

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

September 16, 2024

PART 1 - Problem 1- Solving an ODE using the Laplace Transform

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC]

Using the **Laplace transform** find the solution for the following Ordinary Differential Equation (ODE):

$$\frac{d^2y(t)}{dt^2} = u(t) - m_1 \frac{dy(t)}{dt} - m_2 y(t)$$

with $m_1 = 4$, $m_2 = 8$, and $u(t)$ is the **causal input**

$$u(t) = 15 \sin(16t) \cdot \mathbf{1}(t) - 42 \cdot \mathbf{1}(t - 23)$$

As usual, $\mathbf{1}(t)$ denotes the unit step function: $\mathbf{1}(\tau) = \begin{cases} 0 & \leftrightarrow \tau < 0 \\ 1 & \leftrightarrow \tau \geq 0 \end{cases}$

Finally, consider as **initial conditions** (ICs)

$$\begin{cases} y(0) &= 42 \\ \frac{dy(t)}{dt} \Big|_{t=0} &= -1 \end{cases}$$

Assignment

Your solution has to:

- Use the Matlab's **Symbolic Toolbox**.
- Solve the ODE in the Laplace transform domain, taking into account the assigned initial conditions.
- Evaluate the inverse Laplace transform of the solution, making use of the available Symbolic Toolbox commands and functions .
- Assign the solution in the time domain to the **symbolic** variable **y_sol**.

Solution

Compare with Grader-Laplace-Problem 5

```
syms t s
syms y(t) u(t) % the variables in the time domain
syms Y % the Laplace transform
```

```

assume(t >=0)

% ---- input time function -----
u(t) = 15*sin(16*t)*heaviside(t)-42*heaviside(t-23);

% ----- ICs -----
y0VAL = sym(42);      % the initial conditions: the value y(0)
dotY0VAL = sym(-1);   % the initial conditions: the value dot_y(0)

% ----- the time derivatives of the function y(t) -----
dot_y = diff(y,t); % the 1st derivative of y
ddot_y = diff(dot_y,t);
% -----

% ---- the ODE ----
eqnODE = ddot_y == u -4*dot_y - 8*y; % the ODE to be solved
% ---- the ODE ----

eqnLT = laplace(eqnODE, t, s); % applying Laplace transform

eqnLT1 = subs(eqnLT,laplace(y,t,s),Y);
eqnLT2 = subs(eqnLT1, y(0), y0VAL);
eqnLT3 = subs(eqnLT2, subs(diff(y(t), t), t, 0), dotY0VAL)

```

eqnLT3 =

$$Y s^2 - 42 s + 1 = \frac{240}{s^2 + 256} - 4 Y s - \frac{42 e^{-23 s}}{s} - 8 Y + 168$$

```
Ysol = simplify(solve(eqnLT3, Y))
```

Ysol =

$$\frac{42 s - \frac{42 e^{-23 s}}{s} + \frac{240}{s^2 + 256} + 167}{s^2 + 4 s + 8}$$

```
y_sol = ilaplace(Ysol,s, t)
```

y_sol =

$$42 \operatorname{heaviside}(t - 23) \left(\frac{e^{46 - 2 t} (\cos(2 t - 46) + \sin(2 t - 46))}{8} - \frac{1}{8} \right) - \frac{93 \sin(16 t)}{1640} - \frac{3 \cos(16 t)}{205} + 42 e^{-2 t} (\cos(16 t) + \sin(16 t))$$

